

AK VisionFlow Owner Android App

Phase 3 — Business Alerts & Push Notifications

Phase 3 adds proactive business monitoring. Instead of the owner opening the app repeatedly to check for problems, VisionFlow identifies important changes and sends concise, actionable notifications. The Android app remains strictly read-only.

1. Phase Objective

Build a reliable alert and notification layer that detects important business events, prioritizes them by severity, delivers them to the owner's Android device, and lets the owner open the relevant read-only dashboard view for details.

2. Alert Categories

Category	Examples
Sales	Unusual sales drop, major sales increase, target achieved, significant period-over-period change
Profit	Profit margin drop, unusual profit decline, strong margin improvement
Receivables	Large overdue balance, overdue amount increasing, significant collection change
Inventory	Critical low stock, out-of-stock item, near-expiry/expired stock, dead-stock growth
Expenses	Unusual expense increase or major deviation from normal period
Business Anomaly	Unusual activity or KPI behavior detected by business rules/statistical checks
Performance	Branch/category/product performance changes requiring attention

3. Alert Priority

- **Critical:** Requires prompt owner attention, such as severe sales/profit deterioration, critical stock or major overdue exposure.
- **Important:** Meaningful business issue that should be reviewed soon.
- **Informational:** Positive milestone or useful business update.

4. Push Notification Design

Notifications must be short and useful. The notification should contain the key fact and why it matters, with a deep link into the appropriate read-only screen.

Example:

Profit Margin Alert

Profit margin fell from 34% to 27% this week. Tap to review profit performance.

5. Daily Business Summary Notification

A daily summary can be delivered once per day, preferably at an owner-configured time. It should summarize the most important business movement rather than simply repeat dashboard numbers.

Example:

Daily Business Summary

Sales +12% • Profit +8% • Receivables +19% • 6 low-stock items. Your receivables require the most attention

today.

6. Alert Center Inside the App

- Unread/read state.
- Critical, Important and Informational grouping.
- Date/time of alert.
- Short explanation of the trigger.
- Tap-to-open relevant dashboard/analytics view.
- Optional dismiss/archive behavior without modifying underlying business data.

7. Notification Preferences

Setting	Owner Control
Daily Summary	On / Off
Sales Alerts	On / Off
Profit Alerts	On / Off
Receivable Alerts	On / Off
Inventory Alerts	On / Off
Expense/Anomaly Alerts	On / Off
Minimum Priority	Critical only / Important+ / All

8. Alert Engine Architecture

VisionFlow Data → KPI/Business Rules Engine → Alert Detection → Priority Engine → Notification Service → Android App

Alert detection should run on the backend so alerts are consistent across devices and cannot be bypassed by the client. The Android app receives and displays the result.

9. Duplicate & Noise Control

- Do not send repeated notifications for the same unresolved condition.
- Group related alerts where practical.
- Apply thresholds and minimum-change rules to avoid insignificant alerts.
- Allow the owner to configure notification categories and minimum priority.
- Respect notification permission and device-level notification settings.

10. Deep Linking

Tapping an alert should open the relevant read-only screen, for example: Profit Alert → Profit Analytics; Low Stock → Inventory Visibility; Receivable Alert → Receivables; Sales Alert → Sales Analytics.

11. Backend/API Requirements

API/Service	Purpose
/owner/alerts	Retrieve current and historical owner alerts
/owner/alerts/{id}	Retrieve alert details
/owner/notification-preferences	Read/update notification preferences
Push Token Service	Register and manage Android device push tokens
Alert Engine	Evaluate KPI/business rules and generate prioritized alerts
Notification Service	Deliver push notifications and track delivery state where supported

12. Security

Push notifications must avoid exposing sensitive detailed business/customer information on the lock screen where possible. Detailed information should be retrieved only after secure app authentication. All notification actions remain read-only.

13. Phase 3 Completion Criteria

1. Backend can detect defined business alerts reliably.
2. Alerts have severity/priority and meaningful descriptions.
3. Owner receives Android push notifications for enabled alert categories.
4. Daily business summary notification can be delivered once per day.
5. Notification preferences are configurable.
6. Alert Center displays current and historical alerts.
7. Tapping an alert opens the appropriate read-only dashboard context.
8. Duplicate/noise suppression prevents unnecessary repeated notifications.
9. Read-only security remains enforced.
10. The system is ready for Phase 4 AI Advisor to consume the same business intelligence and alert signals.

14. Phase 3 Result

Result: AK VisionFlow becomes proactive rather than passive. The owner no longer needs to continuously open the dashboard to discover important changes. The system identifies meaningful business events, prioritizes them, and brings them to the owner's phone.

Next Phase: Phase 4 — AI Business Advisor & Daily AI Recommendations.